

SMART CITY HACKATHON LAHORE

Lahore Civic Nervous System

Live air quality and population exposure across Lahore District

Theme 2, City Intelligence. Problem Statement 1: A Real-Time Nervous System for Lahore's Civic Data.

Team Members:

Team Lead: Mahnoor

Email: mahnoorshafiq212@gmail.com

Participant 2: Mahnoor Shabbir

Email: tulipkardo123@gmail.com

Participant 3: Muhammad Awais Asif

Email: m.awaisasif1026@gmail.com

Report date: 3rd September 2026

Live application: lahore-civic-nervous-system.vercel.app

Source code: github.com/Mahnoor9779/Smart_City_Hackathon

Table of Contents

Contents.....	Error! Bookmark not defined.
1. Executive summary	3
2. Problem and approach.....	3
2.1 The gap the brief identifies	3
2.2 The design principle adopted.....	3
3. What was built	3
3.1 Visual and interaction design	4
4. Data sources and verification findings.....	4
4.1 OpenStreetMap has no union council boundaries for Lahore	4
4.2 Two URLs in the brief's source list are incorrect.....	5
4.3 The AQICN demonstration token returns the wrong city	5
4.4 Licensing is a design constraint, not paperwork.....	5
5. Technical implementation.....	5
5.1 Stack	5
5.2 Geometry pipeline	5
5.3 Live data	5
5.4 Why inline SVG rather than a tiled map library.....	6
5.5 Honesty about resolution	6
6. Design system and accessibility	6
6.1 The severity ramp inverts between themes	6
6.2 Accessibility measures.....	6
7. Testing and quality assurance	6
8. Live snapshot	7
9. Known limitations	7
10. Licensing and compliance	7
11. Roadmap	8

1. Executive summary

The Lahore Civic Nervous System is a working web application that shows live air quality across Lahore District, broken down by its five tehsils, on a map drawn from real OpenStreetMap boundaries. It is deployed and publicly reachable.

Its governing principle is a single rule applied everywhere: never show a bare reading. Every value in the interface is paired with the population it affects, because a concentration on its own is a measurement, while a concentration next to the number of people breathing it is information a city can act on.

The brief's hard requirement was to pull two or more live data sources onto a single map for one district or union council. That requirement is met and exceeded: the application combines live air quality from Open-Meteo with administrative geometry and area measurements from OpenStreetMap, and adds population exposure, a 72 hour forecast with health thresholds, and a ranked comparison table.

Verification during development produced findings that materially changed the plan, including the discovery that OpenStreetMap holds no union council boundaries for Lahore, and that two source URLs in the official brief are incorrect. Both are documented in section 4.

2. Problem and approach

2.1 The gap the brief identifies

The brief opens with Rio de Janeiro's operations centre, which achieved its result not by building new infrastructure but by connecting data that already existed across departments. Lahore holds comparable pieces. Suthra Punjab GPS-tracks its collection fleet, Punjab EPA operates air monitors, and PITB is developing One Map Punjab. What is missing is any system that joins them, predicts problems, or accepts verified data from citizens.

The brief also states an honest constraint: none of those official systems are open to outside developers. It therefore asks for a prototype built on genuinely open data, designed so an official feed could be added later without a rebuild.

2.2 The design principle adopted

Most air quality tools in Pakistan present a number. This project treats that as insufficient. The interface pairs every reading with its affected population, so the headline figure reads as a consequence rather than a statistic. This single decision drove the data model, the component design, and the choice of which screens to build.

3. What was built

The application has four screens, each answering a different question. Separating them was deliberate: an earlier single screen version could not orient a first time visitor, who had no way of knowing which place the numbers described.

Screen	Route	Question it answers	Key content	Status
Now	/	How bad is it, and for whom	Headline reading, exposure, locator map	Live
Areas	/areas	Where, and how do they compare	District map, comparison list	Live
Area profile	/areas/[id]	One place in depth	Forecast chart, severity scale	Live
Rankings	/rankings	Which needs attention first	Sortable table, comparison bars	Live

3.1 Visual and interaction design

- A district map drawn from real OpenStreetMap geometry, shaded by current air quality, with each tehsil carrying both its name and its reading.
- A 72 hour forecast chart whose dashed lines are the health thresholds, so the reader sees when the air crosses into a worse band rather than only that it is high.
- Sparklines with a plain language trend word on each area card.
- A severity scale that teaches the colour ramp and marks where the current value sits.
- Comparison bars behind the numbers in the rankings table, scaled against the worst area, with that scaling stated in the footnote.
- A three state theme control: system, light, and dark, with the system default correctly handled.

4. Data sources and verification findings

Every access fact below was tested directly against the live services rather than taken from documentation. Three findings changed the implementation.

4.1 OpenStreetMap has no union council boundaries for Lahore

The brief asks for a union council view. An Overpass query across the district bounding box for every administrative level returned relations at levels 6 and 7 only. There are zero relations at level 8 or 9, so union councils are not mapped in OpenStreetMap for Lahore.

Rather than invent a subdivision, the application uses tehsils, which are the finest unit that reliably exists. The interface says tehsil throughout and prints the administrative level in use in the footer. An interface that implies a precision its data lacks is worse than one that states the limit.

Validation of the resulting geometry: the five tehsils sum to 1,809 square kilometres against a district total of approximately 1,772, a difference inside two percent. That agreement is what confirmed the boundary stitching and map projection were correct.

4.2 Two URLs in the brief's source list are incorrect

- The AQICN path given for Lahore returns HTTP 404. The working address is aqicn.org/city/lahore.
- The WorldPop REST API has moved to hub.worldpop.org/rest/data.

4.3 The AQICN demonstration token returns the wrong city

The publicly documented demo token ignores the submitted query and returns Shanghai data for any city name and any coordinate pair. Testing with it yields plausible looking readings that have no relationship to Lahore. This is a silent failure with no error message, and it would corrupt results without any obvious sign.

4.4 Licensing is a design constraint, not paperwork

Two upstream sources restrict redistribution, and both were checked against their published terms.

- AQICN state that their data cannot be redistributed as cached or archived data. Those readings may therefore be displayed but never exported.
- OpenAQ is conditional, and exposes the condition programmatically: a `redistributionAllowed` flag per licence, readable from their API.

This is why the canonical data record carries a required exportable boolean. The licence question travels with the value rather than being remembered at the export endpoint.

5. Technical implementation

5.1 Stack

Next.js 15 with the App Router and TypeScript in strict mode, CSS Modules with a design token system, hand authored inline SVG for the map and all charts, and Vitest for the test suite. Deployed on Vercel.

5.2 Geometry pipeline

The district boundary was retrieved from OpenStreetMap through Nominatim at 1,629 vertices and simplified using the Douglas-Peucker algorithm to 118 points, a reduction of roughly 93 percent with no visible loss at display size. The five tehsil boundaries came from Overpass, with their constituent outer ways stitched into closed rings. All geometry is projected equirectangular with a latitude correction so the shape is not horizontally stretched.

5.3 Live data

Open-Meteo was selected as the primary source because it requires no API key and returns permissive cross origin headers, which means the application carries real readings from its first render with no account to approve and no secret that could leak. A single request returns all five tehsils. Readings refresh on a ten minute cycle.

5.4 Why inline SVG rather than a tiled map library

The map answers one question, which is where the reader is looking, and it must answer it within the first second on a mid range phone. At 118 points the district outline is approximately 1.5 kilobytes, renders immediately, and requires no tiles, no network request, and no JavaScript in order to appear. A tiled map library earns its weight on a dedicated screen with panning and layer controls, but would cost the one moment that has to feel instantaneous.

5.5 Honesty about resolution

Open-Meteo serves a model grid of approximately 11 kilometres, which is coarser than a tehsil. Every reading is therefore marked as interpolated rather than measured, carries a confidence value below one, and renders through an explicit estimated state in the interface. Presenting a smooth surface that implied per tehsil sensors would be a claim the data cannot support.

6. Design system and accessibility

6.1 The severity ramp inverts between themes

In the light theme severity increases with darkness. In the dark theme it increases with luminance. The invariant preserved is visual weight against the background rather than hue, because a dark purple region disappears against a dark map. Both directions are asserted strictly monotonic by an automated test.

The ramp deliberately avoids the conventional red to green scale, which is the hardest pair to distinguish for approximately one in twelve men. Severity is never carried by colour alone: every band is accompanied by its name in text, and the scale legend states each numeric boundary.

6.2 Accessibility measures

- WCAG 2.2 Level AA contrast, verified by automated test in both themes.
- A sortable table equivalent of the map data, at its own address.
- Full keyboard navigation with a visible focus ring on every interactive element.
- A skip link, landmark regions, and accessible names on the map and every chart.
- Reduced motion respected on every transition and animation.
- Minimum 44 pixel touch targets, designed for a 360 pixel screen first.

7. Testing and quality assurance

The project carries 35 automated tests covering colour contrast, theme token parity, severity ramp ordering, and a set of platform content rules enforced as build gates.

Two defects were found by tests that visual review had missed, which is the strongest argument for having written them.

- A contrast test identified a severity band at 3.20 to 1, below the accessibility threshold, in a colour that had been reviewed by eye many times.
- Mutation testing then revealed a more serious problem: the theme parity test was matching an explanatory code comment instead of the actual CSS rule, and had been comparing the light palette against itself. It passed

vacuously. Stripping comments before parsing fixed it, and a further assertion was added to catch value drift between the two dark theme blocks.

The lesson recorded in the project documentation is that a test which cannot fail is worse than no test, because it produces false confidence. Deliberately breaking the code to confirm the guards fire is now part of the process.

8. Live snapshot

Readings captured 30 August 2026 at 20:38 UTC, which is 01:38 Pakistan Standard Time. Values are PM2.5 in micrograms per cubic metre. All five tehsils fall in the Unhealthy band, defined as 55 to 150.

Tehsil	PM2.5	Population	Area, km2	Band
Raiwind	68.5	700,000	505	Unhealthy
Lahore City	67.5	3,600,000	213	Unhealthy
Lahore Cantonment	67.5	2,800,000	441	Unhealthy
Shalimar	67.5	1,600,000	298	Unhealthy
Model Town	64.5	2,400,000	352	Unhealthy

Population weighted district mean: 66.9 micrograms per cubic metre, approximately 13 times the World Health Organization annual guideline of 5. On this snapshot the entire district population of roughly 11.1 million was breathing air in the Unhealthy band or worse.

The mean is population weighted rather than a simple average, because Raiwind with 700,000 residents should not count equally against Lahore City with 3.6 million.

9. Known limitations

These are recorded openly. A system that claims to do everything invites a reviewer to find the overstatement, which then discredits the rest.

- Sensor scarcity. Lahore has few public ground monitors. Values are interpolated from a model grid and are labelled as estimates with a confidence value, rather than presented as measurements.
- No official feeds. Suthra Punjab, Punjab EPA, and PITB One Map are not open to outside developers. Their adapters are designed and mocked. Nothing in this prototype claims to read them.
- Boundary resolution. Union councils are unavailable in OpenStreetMap for Lahore, so tehsils are used and the interface states which level is in use.
- Population figures are apportioned from the 2023 census district total by relative built up density. They are estimates, not counts, and are labelled as such wherever they appear.
- Alerts, Urdu localisation, road network travel times, and the facility siting optimiser are designed and specified but not yet implemented.

10. Licensing and compliance

The project is dual licensed, which is the standard civic technology arrangement.

- Application code under the MIT licence.
- Published data exports under the Open Database License 1.0, with the full canonical licence text included in the repository.
- Documentation under Creative Commons Attribution 4.0.

Layers derived from the OpenStreetMap road graph constitute a Derivative Database under ODbL, so exports carry its share alike clause. That clause reaches the database and never the software, which is why the application itself remains permissively licensed.

Every dataset, library, font, and model is recorded in the repository's attributions file, as the hackathon Code of Conduct requires. Contributions produced with AI assistance are disclosed there.

11. Roadmap

The architecture was specified in five stages, of which the foundation and the working application are complete. The remaining work is specified in detail and ready to build.

- Ground sensor readings from OpenAQ and, when access permits, Punjab EPA, behind server side route handlers so no key reaches the browser.
- WorldPop population rasters replacing the apportioned estimates.
- Road network travel times, so nearest hospital means minutes by road rather than a straight line, in a city divided by canals and railways.
- A facility siting optimiser identifying where a new clinic would bring the largest underserved population within a fifteen minute threshold.
- Citizen reporting with a verification pipeline, addressing the gap the brief explicitly names.
- Full Urdu localisation with right to left layout.